

Joseph Paxton and Gustave Eiffel

Improved production of cast iron and steel enabled 19th-century architects to build on a new and massive scale, and advances in sheet-glass manufacture led to a vogue for conservatories and greenhouses. In 1837, gardener Joseph Paxton (1803–65) built the largest conservatory in the world for his employer the Duke of Devonshire. He later built the vast, prefabricated Crystal Palace in Hyde Park, London, for the Great Exhibition (1851). Covering some 92,000 square metres (990,000 sq ft), the cast-iron, timber and glass structure (opposite) had no precedents. It took eight months to build and even less time to dismantle.

In 1889, French civil engineer and architect Gustave Eiffel (1832–1923) built a tower in Paris as the centrepiece of the Universal Exposition. Designed with an open lattice to minimize wind resistance, it comprised 18,038 separate components, all made in Eiffel's factory on the city's outskirts and assembled on site. Until the construction of New York's Chrysler Building (1930), the Eiffel Tower was the world's tallest structure.

